

Khoa Nguyen

Research Engineer | AI systems, research infrastructure, evidence-driven ML

Ho Chi Minh City, Vietnam | Open to remote roles in Vietnam

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PROFILE

AI engineer with five accepted or published papers spanning handwritten mathematical expression recognition, molecular generation, and vision-language active learning. Builds Python research tools and contributes reviewed open-source changes, with a research interest in reliable industrial visual inspection.

RESEARCH AND ENGINEERING FOCUS

Research infrastructure **Experiment design and validation** **Cross-modal ML** **Python systems** **Evidence and reproducibility**

SELECTED WORK

Research Ops pre-alpha

Research infrastructure | current

- Built a human-gated research workbench that keeps research records, evidence, and decisions recoverable while a human remains responsible for scientific approval.
- Implemented a FastAPI/PostgreSQL backend with immutable revision snapshots, transactional events, idempotent command receipts, and an SSE web surface; added bounded read-only PDF ingestion with SHA-256 artifact and evidence checks.
- Added revision-pinned manuscript export/compile and guarded experiment/runtime controls. Local acceptance and restore checks cover system plumbing only; provider/GPU execution and productivity benefit remain unmeasured.

LexiChem capstone system

2025-2026 | completed

- Contributed to a text-to-molecule workflow connecting natural-language prompts, multi-model inference, and 2D/3D molecular inspection; documented integrations include RDKit, FastAPI, and Triton.
- Worked on data processing, SMILES/SELFIES conversion, model-optimization support, and backend/API integration. The APWeb-WAIM paper is the extracted research contribution from the larger capstone system.

Bernstein open-source contributions

Public repository | reviewed merges

- Contributed a merged verifier-hardening change binding projection checks to authenticated audit evidence across the run, journal head, trace, span count, and projection digest; added failure, archival, and read-only tests ([PR #3672](#)).
- Contributed a merged MCP change that advertises host effects across 25 tool schemas, propagates them through FastMCP, and checks schema/tier drift ([PR #3673](#)).

SELECTED PUBLICATIONS

Mask CoMER: Enhancing Handwritten Mathematical Expression Recognition with Masked Language Pretraining and Regularization

ICDAR 2025 | Published | Co-first author ([link](#))

ChemAligner-T5: A Unified Text-to-Molecule Model via Representation Alignment

ICCIES 2026 | Published | Co-first author ([link](#))

LexiChem: Shared-Latent Alignment for Faithful Text-to-Molecule Generation in SELFIES Space

APWeb-WAIM 2026 | Accepted | First author; conference presentation completed

CorrTie: Correction-aware Tie-breaking for Active Learning with Vision-Language Models

APWeb-WAIM 2026 | Accepted | Co-author; writing/review contribution

Enhancing handwritten mathematical expression recognition with convolutional block attention refinement module and multi-task learning

Journal of Information and Telecommunication | Published online 19 August 2026 | Co-author ([link](#))

CURRENT RESEARCH

Prefix-Derived Structural Relative Bias (PSRB) for CoMER-based HMER - submitted to FISAT.

Research interest: reliable industrial visual inspection under changing products and limited labels, with language-grounded and active evidence acquisition.

EDUCATION AND RECOGNITION

AI & Data Science, FPT University, Ho Chi Minh City

Completed all academic requirements; degree conferral expected November 2026. Final GPA: 8.0/10.

Recognition: Second Prize, university research festival / RESFES (2025 and 2026).

TOOLS AND METHODS

Python, PyTorch, Transformers, FastAPI, Pydantic, RDKit, SQLite/PostgreSQL, Docker, Git, experimental design, reproducibility, structured evidence capture.